

ECHO MUSIC PLAYER

Project Report

<Template Version 1.3>

PROJECT WORK PHASE – II (ECS899)

Degree

BACHELOR OF TECHNOLOGY (CSE)

PROJECT GUIDE:

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FACULTY OF ENGINEERING AND COMPUTING SCIENCES

TEERTHANKER MAHAVEER UNIVERSITY, MORADABAD

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I would really like to specify my unique gratitude and way to enterprise teachers for giving me such attention and time.

My thanking and appreciations additionally attend my colleagues in suggesting the task and folks who have volitionally helped me out with their skills.

Manish Gupta

Place: Teerthanker Mahaveer University, Moradabad

Date: 25/09/2020

DECLARATION

I hereby declare that this Project Report titled “**ECHO MUSIC PLAYER**” submitted by me and approved by our project guide, to the Faculty of Engineering and Computing Sciences (FOECS), Teerthanker Mahaveer University, Moradabad, is a bonafide work undertaken by me and it is not submitted to any other University or Institution for the award of any degree diploma / certificate or published any time before.

Student Name:	Manish Gupta	Signature
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Project Guide: (External)	Name	Signature
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Project Guide: (Internal)	Mr. Rupal Gupta	Signature
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1 Project Title

Echo Music Player – Android Application

2 Problem Statement

In this project, we aim to build a general purpose Music Player Android application with the following functionalities:

- A glossy Splash screen and a custom Navigation drawer with app logo section at the top along with links to all other fragments of the App.
- An 'All songs' screen, the home screen of the app, mentioning songs fetched from device and external storage media on device.
- The app should be able to fetch and play possibly all sorts of audio files.
- A 'Favorites' screen displaying lists of song marked as favorites earlier.
- A 'Settings' screen displaying app configuration settings.
- An 'About us' screen displaying more information about the project developer.
- A 'Now playing' screen with powerful equalizer and other necessary controls and options.
- A 'Now playing' mini bar at the bottom with name of the track playing and play or pause feature. This would appear if the user has transitioned from 'Now playing' screen to either the 'All songs' screen or the 'Favorites' screen without pausing the track.
- Background play. The app will continue playing the track if the app gets minimized without the music being paused.
- A notification saying "A track is playing in the background" only if the app gets minimized without the music being paused. (depreciated in v1.1 after Google's Policy Update Jan 2019)

With this project, not only I hands on practiced Android App Development skills, but also made a contribution to a significant domain of Computing, It's projected that by 2022, the number of users on Android will be around 859 million, also I published this project on Google Play Store, initially when developed it and now it has attained close to 87k+ downloads on the same, here is the link to it: <https://play.google.com/store/apps/details?id=com.manishgupta1998.echo&hl=en>

3 Project Description

Echo is a simple, yet easy to use Music Player App designed using Open Source Code having a powerful Equalizer. It is Free to use and download. It Sorts the Music on the basis of Name as well as on the basis of Date Added. Also, it supports nearly all Music/Audio File Formats. The Embedded equalizer in the App is what you will love like anything and it also imparts the Professional Look to the App. You can regulate the Equalizer using volume adjustments. Enjoy your Music. You can add music to favorites section in the App Library and play them later. The All New Shake to Change the song Feature will touch your hearts deep and I bet you'll fall in Love with the App.

3.1 Scope of the Work

- Shake to Change Functionality : Give your phone a shake to change the playing track
- Supports All types of Audio Files and is a User Friendly App - with Great UI and UX.
- Powerful attention-hacking Open Source Equalizer in a general purpose music player.
- In-built Headset/Bluetooth Controls with the App.
- The iOS Version haven't been worked upon, but would be done later sometime
- Smart Remembering feature automatically applies previously Selected Controls
- Create Custom Playlists by Adding your Tracks to Favorites

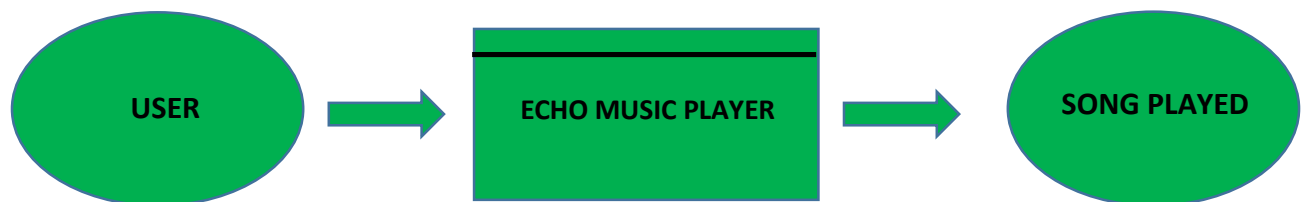
3.2 Project Modules

This Project contains a number of modules, which are sort of inter-connected or independent in nature, however, mainly, the modules of this project can be classified broadly into three types, they are:

- Activities: Activity in an Android App is a major component providing the interactive aspect in the App UI. Here, we have used two major Activities in our Music Player App, the Splash Activity (at Launch) and Main Activity (for remaining in App features).

- **Fragments:** Fragments are the mini panels that we want to be displayed over an activity, its similar to opening different screens in a single window as fragments is to activity is. Here, we are using 5 Fragments in our Music Player App for handling various in App events during runtime, namely mainScreenFragment, AboutUsFragment, SongPlayingFragment etc.
- **Adapters:** An adapter is a bridge that acts, coordinates and controls actions between Activities and Fragments, this App uses 3 Adapters, the FavoritesAdapter, MainScreenAdapter and NavigationDrawerAdapters.

3.3 Context Diagram (High Level)



4 Implementation Methodology

Please See Annexures

5 Technologies to be used

5.1 Software Platform

Front-End

- XML

Back-end

- Kotlin
- SQLite

5.2 Hardware Platform

- a) RAM : at least 256 MB of RAM
- b) Hard Disk : at least 10 MB of Disk Space on Hard Disk
- c) OS : Android Operating System
- d) IDE : Android Studio v1.0

5.3 Tools

Tool Name: Android Studio IDE

Version: 1.0

Developer(s): Google Inc.

Written in: Java, Kotlin, C++

6 Advantages of this Project

- Shake to Change Functionality : Give your phone a shake to change the playing track
- Supports All types of Audio Files
- Powerful attention-hacking Open Source Equalizer
- Headset/Bluetooth Controls
- Create Custom Playlists by Adding your Tracks to Favorites
- Smart Remembering feature automatically applies previously Selected Controls
- User Friendly - Great UI and UX

7 Assumptions, if any

Basically, “Echo – Music Player” application needs a Mobile device running on Android OS v4.0.4 or above and after Installation, it needs access to some permissions like Storage and Audio settings to function desirably, these permissions are optional though recommended as the App may misbehave without granting them or may crash abruptly.

8 Future Scope and further enhancement of the Project

It is easier to extend the App that we have developed in this Project. The App here serves the basic minimum requirement for the Music Player application that may be additionally used on Mobile devices. I at some point of time later can possibly extend the features of this App to Add Quick Song Search, Displaying of Song thumbnail while playing, Notification area control panel while playing could be added and other than that, various themes and personalization schemes could be supplied, also, we can add cloud connectivity to App allowing user for adding latest music to Favourites or download them to device as seen in other such apps in Market offering subscription plans to Users for them allowing In-App purchases, however, a long journey to go, this is just an Initial Prototype of the bare minimum, and would prove a baseline for its further development and other similar product developments from me in the future. Ending this with my favourite thought that “No Software can be ever perfect, there exists scope of improvement in the future”.

9 Project Repository Location

S#	Project Artifacts (softcopy)	Location (Mention Lab-ID, Server ID, Folder Name etc.)	Verified by Project Guide	Verified by Lab In-Charge
1.	Project Synopsis Report (Final Version)		Name and Signature	Name and Signature
2.	Project Progress updates		Name and Signature	Name and Signature
3.	Project Requirement specifications		Name and Signature	Name and Signature
4.	Project Report (Final Version)		Name and Signature	Name and Signature
5.	Test Repository		Name and Signature	Name and Signature
6.	Any other document, give details		Name and Signature	Name and Signature

10 Definitions, Acronyms, and Abbreviations

Abbreviation	Description
DFD	Data Flow Diagram
IDE	Integrated Development Environment
OS	Operating system
RAM	Random Access Memory

11 Conclusion

The Echo Music Player App developed using XML, Kotlin and SQLite fully meets the objectives of the need for which it has been developed. The App has reached a steady state where all possible identified and tested bugs have been eliminated. As so much as perfection is bothered no App is perfect ever and there square measure probabilities of improvement every moment. Further, it fulfils the basic requirement for its existence and would be enhanced in future, it has now attained 87k+ downloads in Google Play Store and is satisfactorily performing there.

12 References

a) Books

- Android Development with Kotlin by Igor Wojda
- Software Engineering Principles and Practices by Roger Pressman

b) Websites

- www.google.com
- www.trainings.internshala.com
- www.youtube.com
- www.edx.com
- www.stackoverflow.com

S#	Reference Details	Owner	Version	Date
1.	Project Synopsis	Manish Gupta	1.0	03-12-20

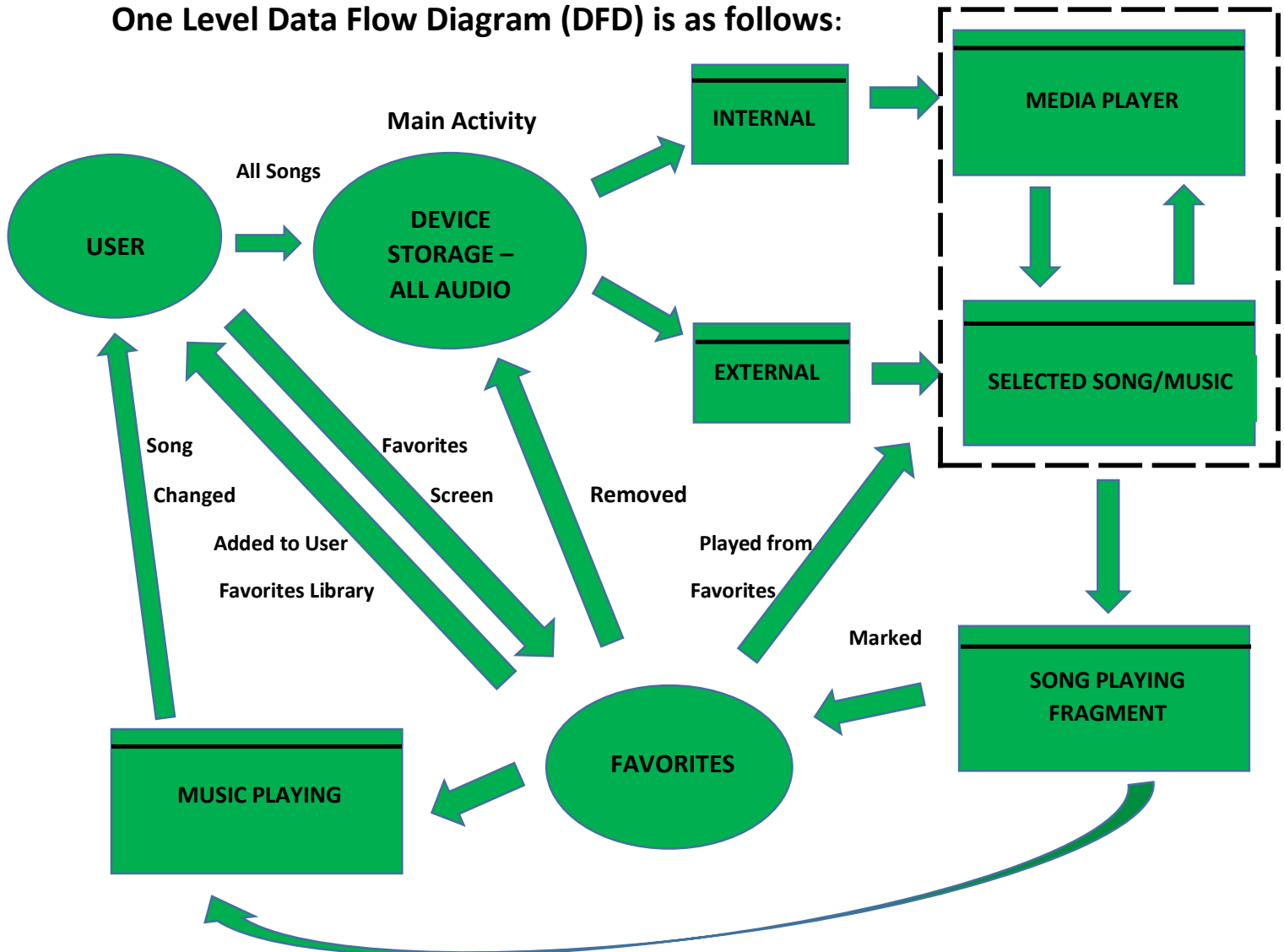
Annexure A

Data Flow Diagram (DFD)

Zero Level Data Flow Diagram (DFD) is as follows:



One Level Data Flow Diagram (DFD) is as follows:



Annexure B

Data Dictionary (DD)

FavoriteTable (in FavoriteDatabase)

Fields	Data type	Description
SongID	Integer	Song ID
SongTitle	Text	Song Name
SongArtist	Text	Artist Name
SongPath	Text	Song's Physical Location

Annexure C Screenshots

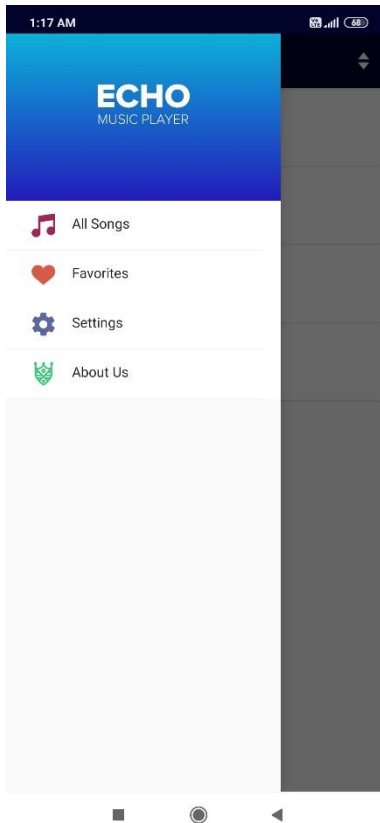
App Icon



Splash Screen, Launch



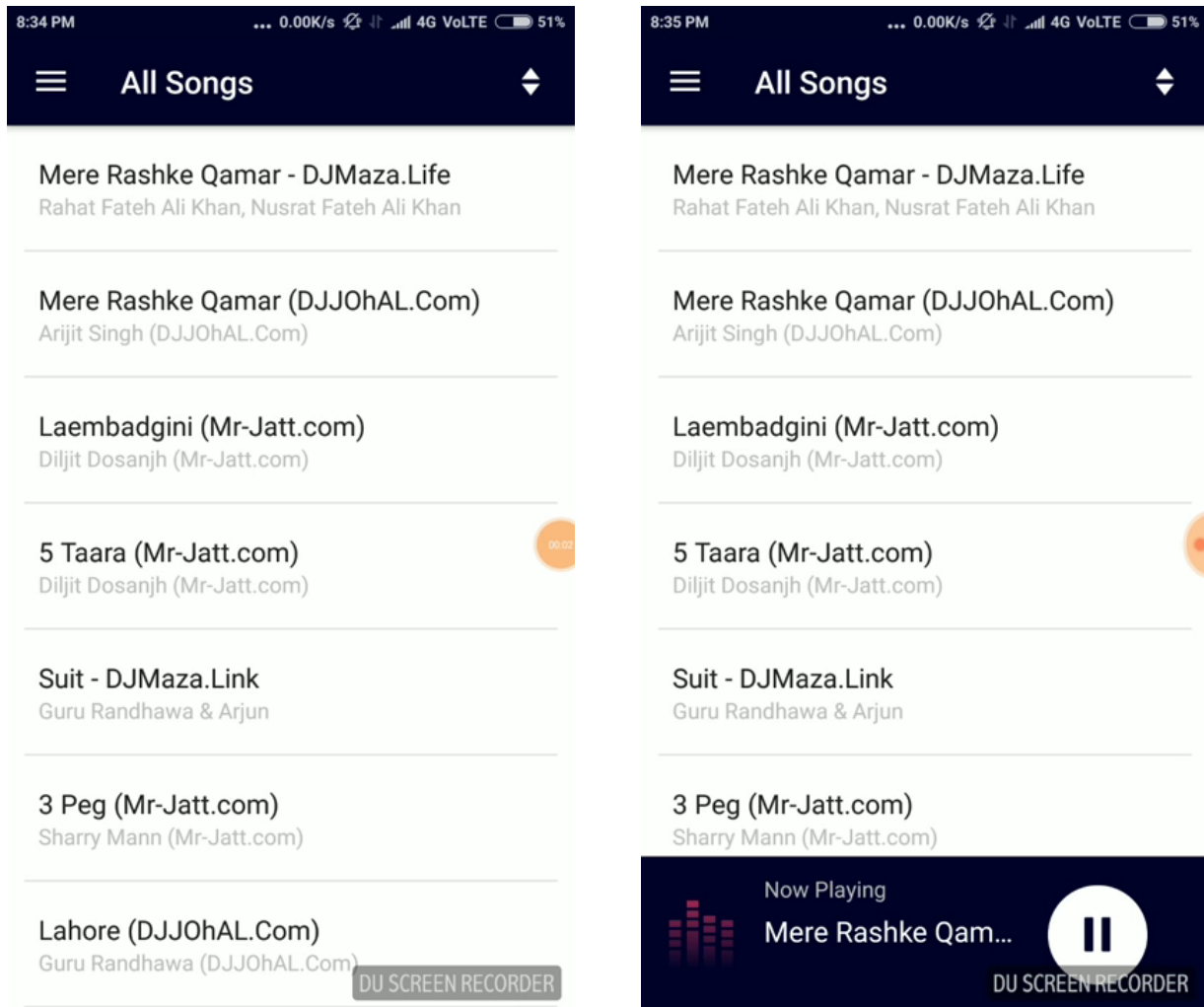
Custom Navigation Screen



Song Playing Screen



All Songs, Song Playing and Favorites Screens:



App Banner:



Song Playing and Favorites Screens:



End of Project Report

